DAY 3 – OVERTHEWIRE.ORG – CTF FOR LEARNING LINUX COMMANDS

The goal of this level is for you to log into the game using SSH. The host to which you need to connect is **bandit.labs.overthewire.org**, on port 2220. The username is **bandit0** and the password is **bandit0**. Once logged in, go to the Level 1 page to find out how to beat Level 1.

SOLN:

C:\Users\nehaa>ssh -p 2220 bandit0@bandit.labs.overthewire.org

The password for the next level is stored in a file called **readme** located in the home directory. Use this password to log into bandit1 using SSH. Whenever you find a password for a level, use SSH (on port 2220) to log into that level and continue the game.

SOLN:

cat readme

PW - 6y2kwnwK6grgvwvpvLaa2T1cpFEKOhNR

The password for the next level is stored in a file called **-** located in the home directory

SOLN:

C:\Users\nehaa>ssh -p 2220 bandit1@bandit.labs.overthewire.org

cat ./"-"

PW - PK8fYLZg2hnHSz83plBL1iEPKdD3QToB

The password for the next level is stored in a file called `--spaces in this filename--` located in the home directory

SOLN:

ssh -p 2220 bandit2@bandit.labs.overthewire.org

cat ./"--spaces in this filename--"

PW - 7ZZ2LFrykP2zEyvBl4m3clcL7tGYJPME

The password for the next level is stored in a hidden file in the **inhere** directory.

```
bandit3@bandit:~$ ls
inhere
bandit3@bandit:~$ cd inhere
bandit3@bandit:~/inhere$ ls
bandit3@bandit:~/inhere$ ls -la
total 12
drwxr-xr-x 2 root    root    4096 Jun 24 14:59 .
drwxr-xr-x 3 root    root    4096 Jun 24 14:59 ..
-rw-r----- 1 bandit4 bandit3   33 Jun 24 14:59 ...Hiding-From-You
bandit3@bandit:~/inhere$ cat ./"...Hiding-From-You"
xzTXq1rDJQVVAzdv5cHq1TQytTWufAMq
bandit3@bandit:~/inhere$ 
```

The password for the next level is stored in the only human-readable file in the **inhere** directory.

```
bandit4@bandit:~$ ls
inhere
bandit4@bandit:~$ cd inhere
bandit4@bandit:~/inhere$ ls -la
total 48
-rw-r----- 1 bandit5 bandit4   33 Jun 24 14:59 -file00
-rw-r----- 1 bandit5 bandit4   33 Jun 24 14:59 -file01
-rw-r----- 1 bandit5 bandit4   33 Jun 24 14:59 -file02
-rw-r----- 1 bandit5 bandit4   33 Jun 24 14:59 -file03
-rw-r----- 1 bandit5 bandit4   33 Jun 24 14:59 -file04
-rw-r----- 1 bandit5 bandit4   33 Jun 24 14:59 -file05
-rw-r----- 1 bandit5 bandit4   33 Jun 24 14:59 -file06
-rw-r----- 1 bandit5 bandit4   33 Jun 24 14:59 -file07
-rw-r----- 1 bandit5 bandit4   33 Jun 24 14:59 -file08
-rw-r----- 1 bandit5 bandit4   33 Jun 24 14:59 -file09
drwxr-xr-x 2 root    root    4096 Jun 24 14:59 .
drwxr-xr-x 3 root    root    4096 Jun 24 14:59 ..
bandit4@bandit:~/inhere$ file ./*
./-file00: data
./-file01: data
./-file02: data
./-file03: data
./-file04: data
./-file05: data
./-file06: OpenPGP Public Key
./-file07: ASCII text
./-file08: data
./-file09: Motorola S-Record; binary data in text format
bandit4@bandit:~/inhere$ cat ./"-file07"
6C7h9GD8M6ai5nr7wo1RonrzFjj9yIrG
bandit4@bandit:~/inhere$
```

The password for the next level is stored in a file somewhere under the **inhere** directory and has all of the following properties:

- human-readable
- 1033 bytes in size
- not executable

```
bandit5@bandit:~$ ls
inhere
bandit5@bandit:~$ cd inhere
bandit5@bandit:~/inhere$ ls -la
total 88
drwxr-x--- 22 root bandit5 4096 Jun 24 14:59 .
drwxr-xr-x  3 root root    4096 Jun 24 14:59 ..
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere00
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere01
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere02
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere03
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere04
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere05
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere06
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere07
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere08
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere09
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere10
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere11
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere12
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere13
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere14
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere15
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere16
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere17
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere18
drwxr-x---  2 root bandit5 4096 Jun 24 14:59 maybehere19
bandit5@bandit:~/inhere$ find type -f size -1033c
find: unknown predicate `-f'
bandit5@bandit:~/inhere$ find -type f -size 1033c
./maybehere07/.file2
bandit5@bandit:~/inhere$ cat "./maybehere07/.file2"
pXa26xhMWaC2SvDotA4r9EgZkulOeSBW
```

The password for the next level is stored **somewhere on the server** and has all of the following properties:

- owned by user bandit7
- owned by group bandit6
- 33 bytes in size

```
bandit6@bandit:~$ find / -user bandit7 -group bandit6 -size 33c 2>/dev/null
/var/lib/dpkg/info/bandit7.password
bandit6@bandit:~$ cat ./"/var/lib/dpkg/info/bandit7.password"
cat: .//var/lib/dpkg/info/bandit7.password: No such file or directory
bandit6@bandit:~$ cat .*"/var/lib/dpkg/info/bandit7.password"
cat: '.*/var/lib/dpkg/info/bandit7.password': No such file or directory
bandit6@bandit:~$ cat "/var/lib/dpkg/info/bandit7.password"
Bmnnvf82KzQlfxgAI2d1zYbr1u9pr3E3
bandit6@bandit:~$
```

The password for the next level is stored in the file **data.txt** next to the word **millionth**

```
bandit7@bandit:~$ ls
data.txt
bandit7@bandit:~$ grep "millionth" data.txt
millionth       VR1ljMayciFxbnUokuQmJFw6QC9VKtub
bandit7@bandit:~$
```

The password for the next level is stored in the file **data.txt** and is the only line of text that occurs only once

```
bandit8@bandit:~$ sort data.txt | uniq -u
EjmOSvuAu7sGAHqHVcBDPirRe9T03kxl
bandit8@bandit:~$
```

The password for the next level is stored in the file **data.txt** in one of the few human-readable strings, preceded by several '=' characters.

```
bandit9@bandit:~$ grep "="
^C
bandit9@bandit:~$ strings data.txt | grep "="
=KGEn
cL0========== the
=<P&
========== password
bU=\
a7=P
>========== is
wbp=
=lR(
a=-io
R========== B0s2khmbT9u0geKuOoVGW3JZKhndE3BG
)=Lc
x=E$
bandit9@bandit:~$
```

The password for the next level is stored in the file **data.txt**, which contains base64 encoded data

```
bandit10@bandit:~$ ls
data.txt
bandit10@bandit:~$ base64 -d data.txt
The password is pYfOY6HwUsDj5rL9UvyhU7MCmv8vN5Ro
bandit10@bandit:~$
```

Decodes the base64 encoded password!

The password for the next level is stored in the file **data.txt**, where all lowercase (a-z) and uppercase (A-Z) letters have been rotated by 13 positions

```
bandit11@bandit:~$ ls
data.txt
bandit11@bandit:~$ cat data.txt | tr 'A-Za-z' 'N-ZA-Mn-za-m'
The password is GROozWPO8QyN0mGrjUkID0WCYkZiQxrN
bandit11@bandit:~$
```

The password for the next level is stored in the file **data.txt**, which is a hexdump of a file that has been repeatedly compressed. For this level it may be useful to create a directory under /tmp in which you can work. Use mkdir with a hard to guess directory name. Or better, use the command “mktemp -d”. Then copy the datafile using cp, and rename it using mv (read the manpages!)

```
bandit12@bandit:/$ mkdir /tmp/trial
bandit12@bandit:/$ cd /tmp/trial
bandit12@bandit:/tmp/trial$ cp ~/data.txt .
bandit12@bandit:/tmp/trial$ ls
data.txt
bandit12@bandit:/tmp/trial$ xxd -r data.txt > data
bandit12@bandit:/tmp/trial$ file data
data: gzip compressed data, was "data2.bin", last modified: Wed Jun 24 14:58:58 2026, max compression, from Unix, original size modul
o 2^32 578
bandit12@bandit:/tmp/trial$ mv data data.gz
bandit12@bandit:/tmp/trial$ gunzip data.gz
bandit12@bandit:/tmp/trial$ file data
data: bzip2 compressed data, block size = 900k
bandit12@bandit:/tmp/trial$ mv data data.bz2
bandit12@bandit:/tmp/trial$ bunzip2 data.bz2
bandit12@bandit:/tmp/trial$ file data
data: gzip compressed data, was "data4.bin", last modified: Wed Jun 24 14:58:58 2026, max compression, from Unix, original size modul
o 2^32 20480
bandit12@bandit:/tmp/trial$ mv data data.gz
bandit12@bandit:/tmp/trial$ gunzip data.gz
bandit12@bandit:/tmp/trial$ file data
data: POSIX tar archive (GNU)
bandit12@bandit:/tmp/trial$ mv data data.tar
bandit12@bandit:/tmp/trial$ tar -xf data.tar
bandit12@bandit:/tmp/trial$ file data
data: cannot open 'data' (No such file or directory)
bandit12@bandit:/tmp/trial$ ls -l
total 36
-rw-rw-r-- 1 bandit12 bandit12 20480 Aug 14 18:20 data.tar
-rw-r----- 1 bandit12 bandit12  2639 Aug 14 18:20 data.txt
-rw-r--r-- 1 bandit12 bandit12 10240 Jun 24 14:58 data5.bin
bandit12@bandit:/tmp/trial$ file data5.bin
data5.bin: POSIX tar archive (GNU)
bandit12@bandit:/tmp/trial$ tar -xf data5.bin
bandit12@bandit:/tmp/trial$ ls -l
total 40
-rw-rw-r-- 1 bandit12 bandit12 20480 Aug 14 18:20 data.tar
```

```
-rw-rw-r-- 1 bandit12 bandit12 20480 Aug 14 18:20 data.tar
-rw-r----- 1 bandit12 bandit12  2639 Aug 14 18:20 data.txt
-rw-r--r-- 1 bandit12 bandit12 10240 Jun 24 14:58 data5.bin
-rw-r--r-- 1 bandit12 bandit12   225 Jun 24 14:58 data6.bin
bandit12@bandit:/tmp/trial$ file data6.bin
data6.bin: bzip2 compressed data, block size = 900k
bandit12@bandit:/tmp/trial$ mv data6.bin data.bz2
bandit12@bandit:/tmp/trial$ bunzip2 data.bz2
bandit12@bandit:/tmp/trial$ ls-l
ls-l: command not found
bandit12@bandit:/tmp/trial$ ls -l
total 48
-rw-r--r-- 1 bandit12 bandit12 10240 Jun 24 14:58 data
-rw-rw-r-- 1 bandit12 bandit12 20480 Aug 14 18:20 data.tar
-rw-r----- 1 bandit12 bandit12  2639 Aug 14 18:20 data.txt
-rw-r--r-- 1 bandit12 bandit12 10240 Jun 24 14:58 data5.bin
bandit12@bandit:/tmp/trial$ file data
data: POSIX tar archive (GNU)
bandit12@bandit:/tmp/trial$ tar -xf data
bandit12@bandit:/tmp/trial$ ls -l
total 52
-rw-r--r-- 1 bandit12 bandit12 10240 Jun 24 14:58 data
-rw-rw-r-- 1 bandit12 bandit12 20480 Aug 14 18:20 data.tar
-rw-r----- 1 bandit12 bandit12  2639 Aug 14 18:20 data.txt
-rw-r--r-- 1 bandit12 bandit12 10240 Jun 24 14:58 data5.bin
-rw-r--r-- 1 bandit12 bandit12    79 Jun 24 14:58 data8.bin
bandit12@bandit:/tmp/trial$ file data8.bin
data8.bin: gzip compressed data, was "data9.bin", last modified: Wed Jun 24 14:58:58 2026, max compression, from Unix, original size
modulo 2^32 49
bandit12@bandit:/tmp/trial$ mv data8.bin data.gz
bandit12@bandit:/tmp/trial$ gunzip data.gz
gzip: data already exists; do you wish to overwrite (y or n)? y
bandit12@bandit:/tmp/trial$ ls -l
total 40
-rw-r--r-- 1 bandit12 bandit12    49 Jun 24 14:58 data
-rw-rw-r-- 1 bandit12 bandit12 20480 Aug 14 18:20 data.tar
-rw-r----- 1 bandit12 bandit12  2639 Aug 14 18:20 data.txt
```

```
bandit12@bandit:/tmp/trial$ ls -l
total 40
-rw-r--r-- 1 bandit12 bandit12    49 Jun 24 14:58 data
-rw-rw-r-- 1 bandit12 bandit12 20480 Aug 14 18:20 data.tar
-rw-r----- 1 bandit12 bandit12  2639 Aug 14 18:20 data.txt
-rw-r--r-- 1 bandit12 bandit12 10240 Jun 24 14:58 data5.bin
bandit12@bandit:/tmp/trial$ file data
data: ASCII text
bandit12@bandit:/tmp/trial$ cat data
The password is qQYQiHOBPR8zR61qxYqX45quvihF2uzk
bandit12@bandit:/tmp/trial$
```

The password for the next level is stored in **/etc/bandit_pass/bandit14 and can only be read by user bandit14**. For this level, you don’t get the next password, but you get a private SSH key that can be used to log into the next level. Look at the commands that logged you into previous bandit levels, and find out how to use the key for this level.
If you need help with this level: a hint file can be found in the home directory.
Make sure to read the error messages as they are informative.

```
bandit13@bandit:~$ ls
HINT  sshkey.private
bandit13@bandit:~$ cat HINT
If you have trouble with this level, note the following:

1) As for all other levels, this level has a website with information:
   https://overthewire.org/wargames/bandit/bandit14.html
2) No, the level is not broken. To verify, see:
   https://status.overthewire.org/
3) The current version of OverTheWire prevents logging in from one
   level to another via localhost. Log out, and see 1)
4) If you get errors, read the error message on your screen.
   We mean it!
bandit13@bandit:~$ cat /etc/bandit_pass/bandit14
cat: /etc/bandit_pass/bandit14: Permission denied
bandit13@bandit:~$ cat sshkey.private
```


```
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAABlwAAAAdzc2gtcn
NhAAAAAwEAAQAAAYEAuCCoxSR6xDCnsm98kqan1x1JF3mfZ6kZa+BoAHetQ/91F4EKTfmE
E+Xs8yVgLhn1YL0TvyVswMzy33OeFJV/KEzef54V8yZo7jFcx+pQlOF6+BFRy+wsyLASV5
XxD/AtafdbVfLZFGTLC53kOFfT0VxokFmnTlIwRyxRJuNAWP9+fkAtbnfqkkixWqg0ZaLr
1fICYam6vb6ilmNfuiGHZyQNHeTkOKZgAaMnQW6bYlRjnkxsNNxk1pj0sT3MUQdLrPCdyh
l8rXUqZ60NZPA6X82/hJNQJR+zVghXrGmTTlU7MAHN7rTdoq4bIQtDShuZZmd5igp5OvVg
tCn4bRM2CFXl8m0P6gNDpR+VYH5f99MwdXyrDUAzuusH7H6CyawRZZF8rFoqONc6HwTJ2h
YJw2uf85ltD3wR8XWzvnBlf2+U9OEmhs01uEYNtNvhq7Gvz/TGI0wR0hEBmsznSns0jShT
cqVsnTMhHaXse3u2D+tbGoC3wOcOyjeX7Skhq63bAAAFiMnVC+zJ1QvsAAAAB3NzaC1yc2
EAAAGBALggqMUkesQwp7JvfJKmp9cdSRd5n2epGWvgaAB3rUP/dReBCk35hBPl7PMlYC4Z
9WC9E78lbMDM8t9znhSVfyhM3n+eFfMmaO4xXMfqUJThevgRUcvsLMiwEleV8Q/wLWn3W1
Xy2RRkywud5DhX09FcaJBZp05SMEcsUSbjQFj/fn5ALW536pJIsVqoNGWi69XyAmGpur2+
opZjX7ohh2ckDR3k5DimYAGjJ0Fum2JUY55MbDTcZNaY9LE9zFEHS6zwncoZfK11KmetDW
TwOl/Nv4STUCUfs1YIV6xpk05VOzABze603aKuGyELQ0obmWZneYoKeTr1YLQp+G0TNghV
5fJtD+oDQ6UflWB+X/fTMHV8qw1AM7rrB+x+gsmsEWWRfKxaKjjXOh8EydoWCcNrn/OZbQ
98EfF1s75wZX9vlPThJobNNbhGDbTb4auxr8/0xiNMEdIRAZrM50p7NI0oU3KlbJ0zIR2l
7Ht7tg/rWxqAt8DnDso3l+0pIaut2wAAAAMBAAEAAAGAEgTUL1LGFtwCFUK2yK05gKIzjH
IRCPZx7+4qj10m3iAqR84PgZj49W+LVDIkqu5MZpaqT4rsjSOhYv+wCSimJH39SjTgxgZM
v36iK0hBcYhtXchoHlIzAcLFUL/yMtKYxyV3UT5uQwIoIq9lbaQerP7jlrjHWDFP2y85k9
oqams6aEWEjKp8kKs/e/U5B3c9qBbCZ+dRyI7W32vDKvZsB0puZC4JrYeOnqpmRY968lD6
```


```
bandit13@bandit:~$ ls
HINT  sshkey.private
bandit13@bandit:~$ ssh -i sshkey.private -p 2220 bandit14@localhost
The authenticity of host '[localhost]:2220 ([127.0.0.1]:2220)' can't be established.
ED25519 key fingerprint is: SHA256:C2ihUBV7ihnV1wUXRb4RrEcLfXC5CXlhmAAM/urerLY
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Could not create directory '/home/bandit13/.ssh' (Permission denied).
Failed to add the host to the list of known hosts (/home/bandit13/.ssh/known_hosts).
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

!!! You are trying to log into this SSH server with a password on port 2220 from localhost.
!!! Connecting from localhost is blocked to conserve resources.
!!! Please log out and log in again.

backend: gibson-0
Received disconnect from 127.0.0.1 port 2220:2: no authentication methods enabled
Disconnected from 127.0.0.1 port 2220
```

DISCONNECTED FROM LOCAL HOST, hence decided to download private key onto my system and use it to gain access directly

```
bandit13@bandit:~$ exit
logout
Connection to bandit.labs.overthewire.org closed.

C:\Users\nehaa>scp -P 2220 bandit13@bandit.labs.overthewire.org:~/sshkey.private .
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit13@bandit.labs.overthewire.org's password:
sshkey.private                                                  100% 2602     4.1KB/s   00:00

C:\Users\nehaa>ssh -i sshkey.private -p 2220 bandit14@bandit.labs.overthewire.org
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
```

The password for the next level can be retrieved by submitting the password of the current level to **port 30000 on localhost**.

PW - pbLYuZtTg4MgaqfJx8jbA9gKKGqM68A7

bandit14@bandit

|

| nc localhost 30000

↓

localhost:30000

|

| send bandit14 password

↓

password for bandit15

After nc (netcat) localhost opens up a cursor. Enter 14’s password there, connection gets established and bandit15 password is retrieved…

```
bandit14@bandit:~$ cat /etc/bandit_pass/bandit14
aaWecNkG4FhxJQxz07uiwzVP6bJiYS65
bandit14@bandit:~$ nc localhost 30000
aaWecNkG4FhxJQxz07uiwzVP6bJiYS65
Correct!
pbLYuZtTg4MgaqfJx8jbA9gKKGqM68A7
```

The password for the next level can be retrieved by submitting the password of the current level to **port 30001 on localhost** using SSL/TLS encryption.

SOLN : kS0Hf0u5HiXFwKMKFqXvPdOTNGGa0X8V

```
bandit15@bandit:~$ openssl s_client -connect localhost:30001
Connecting to 127.0.0.1
CONNECTED(00000003)
Can't use SSL_get_servername
depth=0 CN=SnakeOil
verify error:num=18:self-signed certificate
verify return:1
depth=0 CN=SnakeOil
verify return:1
---
Certificate chain
 0 s:CN=SnakeOil
   i:CN=SnakeOil
   a:PKEY: RSA, 4096 (bit); sigalg: sha256WithRSAEncryption
   v:NotBefore: Jun 10 03:59:50 2024 GMT; NotAfter: Jun  8 03:59:50 2034 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIFBzCCAu+gAwIBAgIUBLz7DBxA0IfojaL/WaJzE6Sbz7cwDQYJKoZIhvcNAQEL
BQAwEzERMA8GA1UEAwwIU25ha2VPaWwwHhcNMjQwNjEwMDM1OTUwWhcNMzQwNjA4
MDM1OTUwWjATMREwDwYDVQQDDAhTbmFrZU9pbDCCAiIwDQYJKoZIhvcNAQEBBQAD
ggIPADCCAgoCggIBANI+P5QXm9Bj21FIPsQqbqZRb5XmSZZJYaam7EIJ16Fxedf+
jXAv4d/FVqiEM4BuSNsNMeBMx2Gq0lAfN33h+RMTjRoMb8yBsZsC063MLfXCk4p+
09gtGP7BS6Iy5XdmfY/fPHvA3JDEScdlDDmd6Lsbdwhv93Q8M6POVO9sv4HuS4t/
jEjr+NhE+Bjr/wDbyg7GL71BP1WPZpQnRE4OzoSrt5+bZVLvODWUFwinB0fLaGRk
GmI0r5EUOUd7HpYyoIQbiNlePGfPpHRKnmdXTTEZEoxeWWAaM1VhPGqfrB/Pnca+
vAJX7iBOb3kHinmfVOScsG/YAUR94wSELeY+UlEWJaELVUntrJ5HeRDiTChiVQ++
wnnjNbepaW6shopybUF3XXfhIb4NvwLWpvoKFXVtcVjlOujF0snVvpE+MRT0wacy
tHtjZs7Ao7GYxDz6H8AdBLKJW67uQon37a4MI260ADFMS+2vEAbNSFP+f6ii5mrB
18cY64ZaF6oU8bjGK7BArDx56bRc3WFyuBIGWAFHEuB948BcshXY7baf5jjzPmgz
mq1zdRthQB31MOM2ii6vuTkheAvKfFf+llH4M9SnES4NSF2hj9NnHga9V08wfhYc
x0W6qu+S8HUdVF+V23yTvUNgz4Q+UoGs4sHSDEsIBFqNvInnpUmtNgcR2L5PAgMB
AAGjUzBRMB0GA1UdDgQWBBTPo8kfze4P9EgxNuyk7+xDGFtAYzAfBgNVHSMEGDAW
gBTPo8kfze4P9EgxNuyk7+xDGFtAYzAPBgNVHRMBAf8EBTADAQH/MA0GCSqGSIb3
DQEBCwUAA4ICAQAKHomtmcGqyiLnhziLe97Mq2+Sul5QgYVwfx/KYOXxv2T8ZmcR
Ae9XFhZT4jsAOUDK1OXx9aZgDGJHJLNEVTe9zWv1ONFfNxEBxQgP7hhmDBWdtj6d
```

```
    Cipher    : TLS_AES_256_GCM_SHA384
    Session-ID: B8B3B627D4B2AD3B2124FA8499D4DBFE1C232985CCEB7B0029225D67919BB7E2
    Session-ID-ctx:
    Resumption PSK: 3DA9D66026A401951A175641C4944E743B6C16C1558444E35A0A351B8DA7E5D104BA4C951C0860ACBAEE18E90B1469EC
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 300 (seconds)
    TLS session ticket:
    0000 - 03 d7 e4 75 c5 91 9e 48-f8 43 79 37 ab 78 7d 9c   ...u...H.Cy7.x}.
    0010 - 5e 04 4e 9a 4b 1d 21 f4-44 1d f3 1a 5f 8c 12 db   ^.N.K.!.D..._...
    0020 - af 2f fd 05 ef 5c 7f fa-c4 45 2a d7 2a 7e d7 22   ./...\...E*.*~."
    0030 - 7c bf 50 aa b1 01 7e a7-10 8b 93 6a 6f db 01 7e   |.P...~....jo..~
    0040 - 57 27 eb 1e 05 ac ce c5-3f 94 38 60 22 80 c2 06   W'......?.8`"...
    0050 - 5b 5a d2 f3 86 a1 c1 f3-d1 a2 8b 47 f1 69 a7 2f   [Z.........G.i./
    0060 - 00 23 56 08 7c a8 da a6-ce 1e 23 f0 5e 79 62 f7   .#V.|.....#.^yb.
    0070 - b4 14 3d 51 7c db 3a b0-ef 7b 28 c7 b1 04 05 fe   ..=Q|.:..{(.....
    0080 - 84 57 14 8d af 0d 05 8d-99 4a 8b 83 52 1d 90 1a   .W.......J..R...
    0090 - c8 8f 69 7d 5d 35 eb c5-29 7d 3a db e3 4d 44 86   ..i}]5..)}:..MD.
    00a0 - 2a b2 b4 cf 41 a8 0a f1-62 fe 10 33 74 e1 8e 26   *...A...b..3t..&
    00b0 - 37 ea d7 2e d2 27 72 81-87 10 7d 26 4f 29 e3 b6   7....'r...}&O)..
    00c0 - 80 4c a2 d2 8c 84 22 28-ee b1 6a 17 14 e8 d8 f1   .L...."(..j.....
    00d0 - 87 f9 f3 b7 4f 19 e2 d6-03 b5 73 3c 0d a7 cb 3b   ....O.....s<...;

    Start Time: 1786734703
    Timeout   : 7200 (sec)
    Verify return code: 18 (self-signed certificate)
    Extended master secret: no
    Max Early Data: 0
---
read R BLOCK
pbLYuZtTg4MgaqfJx8jbA9gKKGqM68A7
Correct!
kS0Hf0u5HiXFwKMKFqXvPdOTNGGa0X8V

closed
```

LOG ON TO BANDIT15, SET UP SSL CONNECTION AT PORT30001, BENEATH SSH DETAILS, CURSOR WAITS FOR BANDIT15 PASSWORD. ONCE ENTERED, IT GIVES PASSWORD FOR BANDIT16.

The credentials for the next level can be retrieved by submitting the password of the current level to **a port on localhost in the range 31000 to 32000**. First find out which of these ports have a server listening on them. Then find out which of those speak SSL/TLS and which don't. There is only 1 server that will give the next credentials, the others will simply send back to you whatever you send to it.

ssh -p 2220 bandit16@bandit.overthewire.org

nmap -p 31000-32000 localhost → note open ports alone

nmap -sV -sC -p 31046,31518,31691,31790,31960 localhost → choose the one which is not "echo"

openssl s_client -connect localhost:31790 –quiet → get the private key for 17, copy it

nano /tmp/bandit17.private

chmod 600 /tmp/bandit17.private

exit

scp -P 2220 bandit16@bandit.labs.overthewire.org:/tmp/bandit17.private . → enter 16's password, 17 private key gets downloaded into my command prompt itself

ssh -i bandit17.private -p 2220 bandit17@bandit.labs.overthewire.org → access to 17

There are 2 files in the homedirectory: **passwords.old and passwords.new**. The password for the next level is in **passwords.new** and is the only line that has been changed between **passwords.old and passwords.new**

cd ..

ls –la

find ~/ passwords.new

cd bandit17

diff passwords.old passwords.new

< old_password

> new_password (format)

PW - OQxXZjELndr90zuhOTDYBEomI0SZITXI

The password for the next level is stored in a file **readme** in the homedirectory. Unfortunately, someone has modified **.bashrc** to log you out when you log in with SSH.

ssh -p 2220 bandit18@bandit.labs.overthewire.org cat readme

The trick is that `.bashrc` logs you out immediately, so you need to execute a command **without starting the normal interactive shell**. SSH will authenticate and execute without .bashrc

PW - KpsOfPkcP7i1FlIExk2QEjyt6dw8dxZI

To gain access to the next level, you should use the setuid binary in the homedirectory. Execute it without arguments to find out how to use it. The password for this level can be found in the usual place (/etc/bandit_pass), after you have used the setuid binary.

bandit19@bandit:~$ ls -la

total 36

drwxr-xr-x 2 root root 4096 Jun 24 14:59 .

drwxr-xr-x 150 root root 4096 Jun 24 15:02 ..

-rw-r--r-- 1 root root 220 Feb 13 2026 .bash_logout

-rw-r--r-- 1 root root 3851 Jun 24 14:50 .bashrc

-rw-r--r-- 1 root root 807 Feb 13 2026 .profile

-rwsr-x--- 1 bandit20 bandit19 14880 Jun 24 14:59 bandit20-do

./bandit20-do cat /etc/bandit_pass/bandit20

PW - 4pIjcunZ0fK2vmp3IwfG8Vf7VhxD6pOA

There is a setuid binary in the homedirectory that does the following: it makes a connection to localhost on the port you specify as a commandline argument. It then reads a line of text from the connection and compares it to the password in the previous level (bandit20). If the password is correct, it will transmit the password for the next level (bandit21).

ssh -p 2220 bandit20@bandit.labs.overthewire.org

bandit20@bandit:~$ ls -la

total 36

drwxr-xr-x   2 root   root   4096 Jun 24 14:59 .

drwxr-xr-x 150 root   root   4096 Jun 24 15:02 ..

-rw-r--r--   1 root   root   220 Feb 13 2026 .bash_logout

-rw-r--r--   1 root   root   3851 Jun 24 14:50 .bashrc

-rw-r--r--   1 root   root   807 Feb 13 2026 .profile

-rwsr-x---   1 bandit21 bandit20 15604 Jun 24 14:59 suconnect

bandit20@bandit:~$ ./suconnect

Usage: ./suconnect <portnumber>

This program will connect to the given port on localhost using TCP. If it receives the correct password from the other side, the next password is transmitted back.

bandit20@bandit:~$ nc -l -p 12345

OPEN ANOTHER COMMAND PROMP WINDOW

ssh -p 2220 bandit20@bandit.labs.overthewire.org

./suconnect 12345

GO BACK TO OLD WINDOW, TYPE IN 20'S PASSWORD, WHICH IS READ AT THE NEW WINDOW, AND 21'S PASSWORD GETS RETURNED!!

PW - bW9kBv5WC3P4yoDyf12LSdGuNz5ka6hY

A program is running automatically at regular intervals from **cron**, the time-based job scheduler. Look in **/etc/cron.d/** for the configuration and see what command is being executed.

bandit21@bandit:~$ cd /etc/cron.d

bandit21@bandit:/etc/cron.d$ ls -la

total 56

drwxr-xr-x   2 root root 4096 Jul 3 16:19 .

drwxr-xr-x 124 root root 12288 Jun 25 12:37 ..

-rw-r--r--   1 root root   102 Nov 5 2025 .placeholder

-r--r-----   1 root root   47 Jun 24 14:59 behemoth4_cleanup

-rw-r--r--   1 root root   127 Jul 3 16:19 clean_tmp

-rw-r--r--   1 root root   120 Jun 24 14:59 cronjob_bandit22

-rw-r--r--   1 root root   122 Jun 24 14:59 cronjob_bandit23

-rw-r--r--   1 root root   120 Jun 24 14:59 cronjob_bandit24

-rw-r--r--   1 root root   188 Feb 13 2026 e2scrub_all

-r--r-----   1 root root   48 Jun 24 15:01 leviathan5_cleanup

-rw-------   1 root root   138 Jun 24 15:01 manpage3_resetpw_job

-rwx------   1 root root   52 Jun 24 15:03 otw-tmp-dir

bandit21@bandit:/etc/cron.d$ cat cronjob_bandit22

@reboot bandit22 /usr/bin/cronjob_bandit22.sh &> /dev/null

* * * * * bandit22 /usr/bin/cronjob_bandit22.sh &> /dev/null

(program runs every minute as bandit22 and stores at cronjob_bandit22.sh)

bandit21@bandit:/etc/cron.d$ cat /usr/bin/cronjob_bandit22.sh

#!/bin/bash (*read the script, find where password is stored!)*

chmod 644 /tmp/t7O6lds9S0RqQh9aMcz6ShpAoZKF7fgv

cat /etc/bandit_pass/bandit22 > /tmp/t7O6lds9S0RqQh9aMcz6ShpAoZKF7fgv

bandit21@bandit:/etc/cron.d$ cat /tmp/t7O6lds9S0RqQh9aMcz6ShpAoZKF7fgv

RYVux2rHEm9tiXHmLFzuR7Vhx6AZQMEz   (password!)

PW - RYVux2rHEm9tiXHmLFzuR7Vhx6AZQMEz

A program is running automatically at regular intervals from **cron**, the time-based job scheduler. Look in ***/etc/cron.d/*** for the configuration and see what command is being executed.

ssh -p 2220 bandit22@bandit.labs.overthewire.org

bandit22@bandit:~$ cd /etc/cron.d

bandit22@bandit:/etc/cron.d$ ls -la

total 56

drwxr-xr-x   2 root root 4096 Jul 3 16:19 .

drwxr-xr-x 124 root root 12288 Jun 25 12:37 ..

-rw-r--r--   1 root root   102 Nov 5 2025 .placeholder

-r--r-----   1 root root   47 Jun 24 14:59 behemoth4_cleanup

-rw-r--r--   1 root root   127 Jul 3 16:19 clean_tmp

-rw-r--r--   1 root root   120 Jun 24 14:59 cronjob_bandit22

-rw-r--r--   1 root root   122 Jun 24 14:59 cronjob_bandit23

-rw-r--r--   1 root root   120 Jun 24 14:59 cronjob_bandit24

-rw-r--r--   1 root root   188 Feb 13 2026 e2scrub_all

-r--r-----   1 root root   48 Jun 24 15:01 leviathan5_cleanup

-rw-------   1 root root   138 Jun 24 15:01 manpage3_resetpw_job

-rwx------   1 root root   52 Jun 24 15:03 otw-tmp-dir

bandit22@bandit:/etc/cron.d$ cat cronjob_bandit23

@reboot bandit23 /usr/bin/cronjob_bandit23.sh &> /dev/null

* * * * * bandit23 /usr/bin/cronjob_bandit23.sh &> /dev/null

bandit22@bandit:/etc/cron.d$ cat /usr/bin/cronjob_bandit23.sh

#!/bin/bash

myname=$(whoami)

mytarget=$(echo I am user $myname | md5sum | cut -d ' ' -f 1)

echo "Copying passwordfile /etc/bandit_pass/$myname to /tmp/$mytarget"

bandit22@bandit:/etc/cron.d$ echo "I am user bandit23" | md5sum   (following the command in cat /usr/bin/cronjob_bandit23.sh)

8ca319486bfbbc3663ea0fbe81326349 -

bandit22@bandit:/etc/cron.d$ cat /tmp/8ca319486bfbbc3663ea0fbe81326349 -

gKXDTAXnIz3OBxiPjRZ2uqutUlPZrBsw

PW - gKXDTAXnIz3OBxiPjRZ2uqutUlPZrBsw

A program is running automatically at regular intervals from **cron**, the time-based job scheduler. Look in **/etc/cron.d/** for the configuration and see what command is being executed.

bandit23@bandit:~$ cd /etc/cron.d

bandit23@bandit:/etc/cron.d$ ls -la

total 56

drwxr-xr-x   2 root root 4096 Jul 3 16:19 .

drwxr-xr-x 124 root root 12288 Jun 25 12:37 ..

-rw-r--r--   1 root root   102 Nov 5 2025 .placeholder

-r--r-----   1 root root   47 Jun 24 14:59 behemoth4_cleanup

-rw-r--r--   1 root root   127 Jul 3 16:19 clean_tmp

-rw-r--r--   1 root root   120 Jun 24 14:59 cronjob_bandit22

-rw-r--r--   1 root root   122 Jun 24 14:59 cronjob_bandit23

-rw-r--r--   1 root root   120 Jun 24 14:59 cronjob_bandit24

-rw-r--r--   1 root root   188 Feb 13 2026 e2scrub_all

-r--r-----   1 root root   48 Jun 24 15:01 leviathan5_cleanup

-rw-------   1 root root   138 Jun 24 15:01 manpage3_resetpw_job

-rwx------   1 root root   52 Jun 24 15:03 otw-tmp-dir

bandit23@bandit:/etc/cron.d$ cat cronjob_bandit24

@reboot bandit24 /usr/bin/cronjob_bandit24.sh &> /dev/null

* * * * * bandit24 /usr/bin/cronjob_bandit24.sh &> /dev/null

bandit23@bandit:/etc/cron.d$ cat /usr/bin/cronjob_bandit24.sh

```
#!/bin/bash
shopt -s nullglob
myname=$(whoami)
cd /var/spool/"$myname"/foo || exit
echo "Executing and deleting all scripts in /var/spool/$myname/foo:"
for i in * .*;
do
    if [ "$i" != "." ] && [ "$i" != ".." ];
    then
        echo "Handling $i"
        owner="$(stat --format "%U" "./$i")"
        if [ "${owner}" = "bandit23" ] && [ -f "$i" ]; then
            timeout -s 9 60 "./$i"
        fi
        rm -rf "./$i"
    fi
done
```

bandit23@bandit:/tmp$ mkdir neha24

bandit23@bandit:/tmp$ cd /tmp/neha24

bandit23@bandit:/tmp/neha24$ printf '#!/bin/bash\ncat /etc/bandit_pass/bandit24 > /tmp/neha24/password\n' > getpass.sh

bandit23@bandit:/tmp/neha24$ cat getpass.sh

#!/bin/bash

cat /etc/bandit_pass/bandit24 > /tmp/neha24/password

bandit23@bandit:/tmp/neha24$ ls -l /tmp/neha24/

total 4

-rwxrwxr-x 1 bandit23 bandit23 65 Aug 15 18:21 getpass.sh

bandit23@bandit:/tmp/neha24$ chmod 777 /tmp/neha24 (make dir writeable)

bandit23@bandit:/tmp/neha24$ ls -ld /tmp/neha24 (check)

drwxrwxrwx 2 bandit23 bandit23 60 Aug 15 18:21 /tmp/neha24

bandit23@bandit:/tmp/neha24$ cp getpass.sh /var/spool/bandit24/foo/ (copy script)

bandit23@bandit:/tmp/neha24$ ls -l /tmp/neha24/

total 8

-rwxrwxr-x 1 bandit23 bandit23 65 Aug 15 18:21 getpass.sh

-rw-rw-r-- 1 bandit24 bandit24 33 Aug 15 18:27 password

(after 1 min, cron job stores password)

bandit23@bandit:/tmp/neha24$ cat password

hVQMk3lJNsmQ7VF3ubyrNNBom7BOgVXv

PW - hVQMk3lJNsmQ7VF3ubyrNNBom7BOgVXv

A daemon is listening on port 30002 and will give you the password for bandit25 if given the password for bandit24 and a secret numeric 4-digit pincode. There is no way to retrieve the pincode except by going through all of the 10000 combinations, called brute-forcing.
You do not need to create new connections each time

C:\Users\nehaa>ssh -p 2220 bandit24@bandit.labs.overthewire.org

bandit24@bandit:~$ nc localhost 30002

I am the pincode checker for user bandit25. Please enter the password for user bandit24 and the secret pincode on a single line, separated by a space.

Ctrl + C

bandit24@bandit:~$ mktemp -d

/tmp/tmp.Ztv1OyMIj7

bandit24@bandit:~$ cd /tmp/tmp.Ztv1OyMIj7

bandit24@bandit:/tmp/tmp.Ztv1OyMIj7$ nano brute.py

```
{
import socket

password = "hVQMk3lJNsmQ7VF3ubyrNNBom7BOgVXv"

s = socket.create_connection(("localhost", 30002))

print(s.recv(4096).decode())

for i in range(10000):
    pin = f"{i:04d}"
    message = f"{password} {pin}\n"
    s.sendall(message.encode())

    response = s.recv(4096).decode()

    if "Wrong" not in response:
        print("FOUND!")
        print("PIN:", pin)
        print(response)
        break

s.close()
}
```

bandit24@bandit:/tmp/tmp.Ztv1OyMIj7$ python3 brute.py

I am the pincode checker for user bandit25. Please enter the password for user bandit24 and the secret pincode on a single line, separated by a space.

FOUND!

PIN: 3976

Correct!

The password of user bandit25 is SoHfqMOEqIX2IYKVciZxvgpR9a2Djx4P

PW - SoHfqMOEqIX2IYKVciZxvgpR9a2Djx4P

Logging in to bandit26 from bandit25 should be fairly easy… The shell for user bandit26 is not **/bin/bash**, but something else. Find out what it is, how it works and how to break out of it.

C:\Users\nehaa>ssh -p 2220 bandit25@bandit.labs.overthewire.org

bandit25@bandit:~$ ls -la

total 40

drwxr-xr-x 2 root root 4096 Jun 24 14:59 .

drwxr-xr-x 150 root root 4096 Jun 24 15:02 ..

-rw-r----- 1 bandit25 bandit25 33 Jun 24 14:59 .bandit24.password

-rw-r----- 1 bandit25 bandit25 151 Jun 24 14:59 .banner

-rw-r--r-- 1 root root 220 Feb 13 2026 .bash_logout

-rw-r--r-- 1 root root 3851 Jun 24 14:50 .bashrc

-rw-r----- 1 bandit25 bandit25 66 Jun 24 14:59 .flag

-rw-r----- 1 bandit25 bandit25 4 Jun 24 14:59 .pin

-rw-r--r-- 1 root root 807 Feb 13 2026 .profile

-r-------- 1 bandit25 bandit25 2602 Jun 24 14:59 bandit26.sshkey

bandit25@bandit:~$ exit

C:\Users\nehaa>scp -P 2220 bandit25@bandit.labs.overthewire.org:~/bandit26.sshkey C:\Users\nehaa\bandit26.sshkey

NORMAL SSH CONNECTIONS KEPT GETTING DISCONNECTED, SO, MANIPULATING USING VI EDITOR!

SHORTEN THE COMMAND PROMPT WINDOW TO SMALL WINDOW, SHOWING ONLY 3-4 LINES AND THEN RUN:

C:\Users\nehaa>ssh -i C:\Users\nehaa\bandit26.sshkey -p 2220 bandit26@bandit.labs.overthewire.org

When –More—is seen, press "v", then type : set shell=/bin/bash then press ENTER, then type :shell TO CREATE YOUR OWN SHELL!

You should get: bandit26@bandit:~$

Then: bandit26@bandit:~$ cat /etc/bandit_pass/bandit26 jHdv2ELQhT22BkprMNDjybZDAkw1zeBJ

PW - jHdv2ELQhT22BkprMNDjybZDAkw1zeBJ

Good job getting a shell! Now hurry and grab the password for bandit27!

SHORTEN THE COMMAND PROMPT WINDOW TO SMALL WINDOW, SHOWING ONLY 3-4 LINES AND THEN RUN:

C:\Users\nehaa>ssh -i C:\Users\nehaa\bandit26.sshkey -p 2220 bandit26@bandit.labs.overthewire.org

When –More—is seen, press "v", then type : set shell=/bin/bash then press ENTER, then type :shell TO CREATE YOUR OWN SHELL!

You should get: bandit26@bandit:~$

THEN :

bandit26@bandit:~$ ls -la

drwxr-xr-x 3 root root 4096 Jun 24 14:59 .

drwxr-xr-x 150 root root 4096 Jun 24 15:02 ..

-rw-r--r-- 1 root root 220 Feb 13 2026 .bash_logout

-rw-r--r-- 1 root root 3851 Jun 24 14:50 .bashrc

-rw-r--r-- 1 root root 807 Feb 13 2026 .profile

drwxr-xr-x 2 root root 4096 Jun 24 14:59 .ssh

-rwsr-x--- 1 bandit27 bandit26 14880 Jun 24 14:59 bandit27-do *(setuid enabled)*

-rw-r----- 1 bandit26 bandit26 258 Jun 24 14:59 text.txt

bandit26@bandit:~$ ./bandit27-do

Run a command as another user.

(cat bandit27-do gave a non-human-readable file!, so simply running with user)

Example: ./bandit27-do id

bandit26@bandit:~$ ./bandit27-do whoami

bandit27

bandit26@bandit:~$ ./bandit27-do cat /etc/bandit_pass/bandit27

STJLJBRRphMxKB392CT4iOr5CbzPU9ER

PW - STJLJBRRphMxKB392CT4iOr5CbzPU9ER

- bandit27-do is a **setuid executable** owned by bandit27, so it can run commands with bandit27's privileges.
- We used ./bandit27-do cat /etc/bandit_pass/bandit27 to make it read the password file that bandit26 normally cannot access.
- The command therefore revealed the **Bandit 27 password**.

There is a git repository at ssh://bandit27-git@bandit.labs.overthewire.org/home/bandit27-git/repo via the port 2220. The password for the user bandit27-git is the same as for the user bandit27.

OPEN GIT

```
neh@ndomain MINGW64 ~
$ git clone ssh://bandit27-git@bandit.labs.overthewire.org:2220/home/bandit27-
git/repo
Cloning into 'repo'...
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit27-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

There is a git repository at ssh://bandit28-git@bandit.labs.overthewire.org/home/bandit28-git/repo via the port 2220. The password for the user bandit28-git is the same as for the user bandit28.

SINCE ALREADY CLONED GO INTO DIRECTORY IN SEARCH OF ANY FILE STORING BANDIT28'S PASSWORD…

```
neh@ndomain MINGW64 ~
$ cd repo

neh@ndomain MINGW64 ~/repo (master)
$ ls -la
total 17
drwxr-xr-x 1 neh 197609  0 Aug 16 02:08 ./
drwxr-xr-x 1 neh 197609  0 Aug 16 02:07 ../
drwxr-xr-x 1 neh 197609  0 Aug 16 02:08 .git/
-rw-r--r-- 1 neh 197609 69 Aug 16 02:08 README

neh@ndomain MINGW64 ~/repo (master)
$ cat README
The password to the next level is: y8Yd2ssKcpHpud7UvOSOxwamRMzIGIeQ

neh@ndomain MINGW64 ~/repo (master)
$ |
```

PW - `y8Yd2ssKcpHpud7UvOSOxwamRMzIGIeQ`

MINGW64:/c/Users/nehaa

```
neh@ndomain MINGW64 ~
$ cd

neh@ndomain MINGW64 ~
$ mv repo repo27

neh@ndomain MINGW64 ~
$ git clone ssh://bandit28-git@bandit.labs.overthewire.org:2220/home/bandit28-git/repo
Cloning into 'repo'...
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit28-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 9 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), 800 bytes | 66.00 KiB/s, done.
Resolving deltas: 100% (2/2), done.
```

There is a git repository at ssh://bandit29-git@bandit.labs.overthewire.org/home/bandit29-git/repo via the port 2220. The password for the user bandit29-git is the same as for the user bandit29.

```
neh@ndomain MINGW64 ~
$ cd repo

neh@ndomain MINGW64 ~/repo (master)
$ ls -la
total 17
drwxr-xr-x 1 neh 197609   0 Aug 16 02:22 ./
drwxr-xr-x 1 neh 197609   0 Aug 16 02:21 ../
drwxr-xr-x 1 neh 197609   0 Aug 16 02:22 .git/
-rw-r--r-- 1 neh 197609 119 Aug 16 02:22 README.md

neh@ndomain MINGW64 ~/repo (master)
$ cat README.md
# Bandit Notes
Some notes for level29 of bandit.

## credentials

- username: bandit29
- password: xxxxxxxxxx


neh@ndomain MINGW64 ~/repo (master)
$ git log
commit e2e1de5396037bafb23e9bb37c12ebea9b911cfd (HEAD -> master, origin/master, origin/HEAD)
Author: Morla Porla <morla@overthewire.org>
Date:   Wed Jun 24 14:59:20 2026 +0000

    fix info leak

commit 2678cfadd8f2a347bc23e1ea491f702e5b184709
Author: Morla Porla <morla@overthewire.org>
Date:   Wed Jun 24 14:59:20 2026 +0000

    add missing data

commit 9530d526c22b9e6e6ae11070ef8ff8ee21eb2e02
Author: Ben Dover <noone@overthewire.org>
Date:   Wed Jun 24 14:59:20 2026 +0000

    initial commit of README.md

neh@ndomain MINGW64 ~/repo (master)
$ git show ~HEAD1
fatal: ambiguous argument '~HEAD1': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'
```

AS PASSWORD IS CLASSIFIED, WE TAKE GIT LOG AND HEAD TO VIEW PAST COMMITS BEFORE THEY WERE CHANGED!!!!!!!!!!

```
neh@ndomain MINGW64 ~/repo (master)
$ git show HEAD~1
commit 2678cfadd8f2a347bc23e1ea491f702e5b184709
Author: Morla Porla <morla@overthewire.org>
Date:   Wed Jun 24 14:59:20 2026 +0000

    add missing data

diff --git a/README.md b/README.md
index 7ba2d2f..42331d9 100644
--- a/README.md
+++ b/README.md
@@ -4,5 +4,5 @@ Some notes for level29 of bandit.
 ## credentials

 - username: bandit29
-- password: <TBD>
+- password: Em7eGtqaMySwNFjCpwzzHhLhospOcdt0
```

PW - `Em7eGtqaMySwNFjCpwzzHhLhospOcdt0`

```
neh@ndomain MINGW64 ~/repo (master)
$ cd ..

neh@ndomain MINGW64 ~
$ mv repo repo28

neh@ndomain MINGW64 ~
$ git clone ssh://bandit29-git@bandit.labs.overthewire.org:2220/home/bandit29-git/repo
Cloning into 'repo'...
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit29-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 16, done.
remote: Counting objects: 100% (16/16), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 16 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (16/16), done.
Resolving deltas: 100% (2/2), done.
```

There is a git repository at ssh://bandit30-git@bandit.labs.overthewire.org/home/bandit30-git/repo via the port 2220. The password for the user bandit30-git is the same as for the user bandit30.

(when trying to get 30's pw from 29)

```
neh@ndomain MINGW64 ~
$ cd repo

neh@ndomain MINGW64 ~/repo (master)
$ ls -la
total 17
drwxr-xr-x 1 neh 197609   0 Aug 16 02:29 ./
drwxr-xr-x 1 neh 197609   0 Aug 16 02:28 ../
drwxr-xr-x 1 neh 197609   0 Aug 16 02:29 .git/
-rw-r--r-- 1 neh 197609 139 Aug 16 02:29 README.md

neh@ndomain MINGW64 ~/repo (master)
$ cat README.md
# Bandit Notes
Some notes for bandit30 of bandit.

## credentials

- username: bandit30
- password: <no passwords in production!>
```

SO, TAKE GIT BRANCH! The password might have been accidentally left in another branch. Show me ALL branches that exist in this repository (git branch –a)

If we discover a dev branch, we want to look at its README **without switching branches**.

```
neh@ndomain MINGW64 ~/repo (master)
$ git branch -a
* master
  remotes/origin/HEAD -> origin/master
  remotes/origin/dev
  remotes/origin/master
  remotes/origin/sploits-dev

neh@ndomain MINGW64 ~/repo (master)
$ git show origin/dev:README.md
# Bandit Notes
Some notes for bandit30 of bandit.

## credentials

- username: bandit30
- password: jq9Dfg2rXsfYsWMgFuKlXhphjdH7USgX


neh@ndomain MINGW64 ~/repo (master)
$
```

PW - jq9Dfg2rXsfYsWMgFuKlXhphjdH7USgX

MINGW64:/c/Users/nehaa

```
neh@ndomain MINGW64 ~
$ mv repo repo29

neh@ndomain MINGW64 ~
$ git clone ssh://bandit30-git@bandit.labs.overthewire.org:2220/home/bandit30-git/repo
Cloning into 'repo'...
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit30-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.

neh@ndomain MINGW64 ~
$ |
```

There is a git repository at ssh://bandit31-git@bandit.labs.overthewire.org/home/bandit31-git/repo via the port 2220. The password for the user bandit31-git is the same as for the user bandit31.

```
neh@ndomain MINGW64 ~
$ cd repo

neh@ndomain MINGW64 ~/repo (master)
$ ls -la
total 17
drwxr-xr-x 1 neh 197609  0 Aug 16 14:18 ./
drwxr-xr-x 1 neh 197609  0 Aug 16 14:17 ../
drwxr-xr-x 1 neh 197609  0 Aug 16 14:18 .git/
-rw-r--r-- 1 neh 197609 31 Aug 16 14:18 README.md

neh@ndomain MINGW64 ~/repo (master)
$ cat README.md
just an epmty file... muahaha

neh@ndomain MINGW64 ~/repo (master)
$ git log
commit 929c564cd34ca667773e2eb02f74b514bc4eeebf (HEAD -> master, origin/master, origin/HEAD)
Author: Ben Dover <noone@overthewire.org>
Date:   Wed Jun 24 14:59:25 2026 +0000

    initial commit of README.md

neh@ndomain MINGW64 ~/repo (master)
$ git show ~HEAD1
fatal: ambiguous argument '~HEAD1': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'

neh@ndomain MINGW64 ~/repo (master)
$ git tag
secret

neh@ndomain MINGW64 ~/repo (master)
$ git show secret
82NkymblpGBYmIXG6ZQ8YldBYstHpfUf
```

PW - `82NkymblpGBYmIXG6ZQ8YldBYstHpfUf`

A Git tag creates a permanent, human-readable bookmark that points to a specific commit in your repository's history. Unlike branches, which automatically move forward as you add new commits, **tags are completely static and never change**. They are most commonly used to label software release versions and important project milestones.

```
neh@ndomain MINGW64 ~/repo (master)
$ cd ..

neh@ndomain MINGW64 ~
$ mv repo repo230

neh@ndomain MINGW64 ~
$ git clone ssh://bandit31-git@bandit.labs.overthewire.org:2220/home/bandit31-git/repo
Cloning into 'repo'...
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit31-git@bandit.labs.overthewire.org's password:
Permission denied, please try again.
bandit31-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.
```

```
neh@ndomain MINGW64 ~
$ cd repo

neh@ndomain MINGW64 ~/repo (master)
$ ls -la
total 18
drwxr-xr-x 1 neh 197609   0 Aug 16 14:26 ./
drwxr-xr-x 1 neh 197609   0 Aug 16 14:24 ../
drwxr-xr-x 1 neh 197609   0 Aug 16 14:26 .git/
-rw-r--r-- 1 neh 197609   7 Aug 16 14:26 .gitignore
-rw-r--r-- 1 neh 197609 154 Aug 16 14:26 README.md

neh@ndomain MINGW64 ~/repo (master)
$ cat README.md
This time your task is to push a file to the remote repository.

Details:
    File name: key.txt
    Content: 'May I come in?'
    Branch: master
```

Follow the README exactly.

```
neh@ndomain MINGW64 ~/repo (master)
$ echo "May I come in?" > key.txt

neh@ndomain MINGW64 ~/repo (master)
$ cat key.txt
May I come in?
neh@ndomain MINGW64 ~/repo (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean

neh@ndomain MINGW64 ~/repo (master)
$ git add -f key.txt
warning: in the working copy of 'key.txt', LF will be replaced by CRLF the next
time Git touches it                (FORCE COMMIT)

neh@ndomain MINGW64 ~/repo (master)
$ git commit -m "Add key"
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'neh@ndomain.(none)')


neh@ndomain MINGW64 ~/repo (master)
$ git config user.name "bandit31"

neh@ndomain MINGW64 ~/repo (master)
$ git config user.email "bandit31@bandit.labs.overthewire.org"

neh@ndomain MINGW64 ~/repo (master)
$ git status
```

```
On branch master
Your branch is up to date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   key.txt


neh@ndomain MINGW64 ~/repo (master)
$ git commit -m "Add key"
[master 3918e31] Add key
 1 file changed, 1 insertion(+)
 create mode 100644 key.txt

neh@ndomain MINGW64 ~/repo (master)
$ git push origin master

backend: gibson-0
bandit31-git@bandit.labs.overthewire.org's password:
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 14 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 327 bytes | 163.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: ### Attempting to validate files... ####
remote:
remote: .oOo.oOo.oOo.oOo.oOo.oOo.oOo.oOo.oOo.oOo.
remote:
remote: Well done! Here is the password for the next level:
remote: pWuj5jBQ6IgV0NXwiH6g1pXRF8S1YvbT
remote:
remote: .oOo.oOo.oOo.oOo.oOo.oOo.oOo.oOo.oOo.oOo.
remote:
To ssh://bandit.labs.overthewire.org:2220/home/bandit31-git/repo
```

README.md told us exactly what the server wanted:

- Filename: key.txt
- Content: May I come in?
- Branch: master

Then created the required file, and discovered .gitignore was disabling file from being committed. Then, I forced Git to add the file.

Then I tried to commit with (git commit -m "Add key"), which failed because user id was unknown. After configuring username and user's email id, tried commiting again, after which I pushed the commit to the bandit server (git push origin master)

The challenge is designed so that the **remote repository checks what we pushed**. When we successfully push the correct `key.txt`, the server responds with the **password for the next level**.

PW - pWuj5jBQ6IgV0NXwiH6g1pXRF8S1YvbT

END OF BANDIT LAB FOR NOW

CONCLUSION:

The OverTheWire Bandit CTF provided a practical foundation in Linux and cybersecurity by requiring hands-on work with file systems, permissions, SSH, restricted shells, SUID binaries, encoding, compression, binary analysis, and Git. More importantly, it developed an investigative mindset: inspect the environment, understand the evidence, choose the appropriate tool, interpret errors, and iteratively test solutions. Completing Bandit strengthened my Linux command-line skills and provided a strong foundation for progressing toward penetration testing, digital forensics, network security, and more advanced CTF challenges.